Student Name: jkd

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Student ID: 321

1. Define modulation and explain its purpose in wireless communication.

Answer: Modulation is the process of varying a carrier signal's properties, such as amplitude, frequency, or phase, to encode information for transmission. This process is essential for efficient use of the electromagnetic spectrum and minimizing interference between channels.

Score: 0/0

Justification: N/A

Feedback: No feedback provided

2. Calculate the path loss at a distance of 500 meters for a signal frequency of 2 GHz in free space.

Answer: The FSLP formula is dB = 20 log10(d) + 20 log10(f) - 147.55

FSLP(dB) = 20 log10(500) + 20 log10(2) - 147.55

= 20 2.69897 + 20 9.30103 - 147.55

= 53.98 + 186.02 - 147.55

= 92.45 dB

Score: 10.0/10

Justification: The student demonstrates a clear understanding of path loss and its calculation, with no errors or misunderstandings. They correctly apply the Free-Space Path Loss formula and provide well-organized and easy-to-follow calculations. However, the student's answer lacks an explanation of the significance of path loss in wireless communication systems.

Feedback: The student should provide a brief explanation of why path loss is important in wireless communication systems, such as how it affects signal strength and reliability.

Summary:

Total Score: 10.0

Percentage: 50.00%

Grade: F

Overall Feedback: The student's response would benefit from a brief explanation of why path loss is important in wireless communication systems. Specifically, they should discuss how path loss affects signal strength and reliability, providing a clear understanding of its significance in wireless communication systems.